

Progressive Landmark Evaluation in A*: An Empirical Study of Nested Differential Heuristics

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Abstract

Differential landmarks give strong admissible estimates but require many table reads. Lazy A* avoids some of this work by evaluating the full heuristic only when a state becomes competitive in OPEN. We ask whether a fixed intermediate prefix can avoid still more landmark work: states first receive Manhattan distance, then four landmark terms, and only then the remaining terms of a 32-landmark heuristic. We compare Manhattan, eager full-landmark, ordinary lazy full-landmark, and staged evaluation in a shared optimal graph-search implementation. The map-disjoint evaluation contains 800 queries over eight Moving AI maps from four topology families, with one warm-up and eight position-balanced timed repetitions per method and query. All returned costs matched independent BFS, and eager, lazy, and staged full landmark search had identical expansion traces on all 800 queries. Staging saved a median 3.10% of pivot evaluations and 3.02% of distance-table reads relative to ordinary lazy evaluation, but its primary median paired time ratio was 1.0543 (staged/lazy; hierarchical 95% bootstrap interval [0.9890, 1.1016]). Its descriptive bootstrap probability of a ratio below one was 0.053. The crossover was strongly regime-dependent: both maze maps favored staging, whereas both warehouse maps had zero median read saving and favored ordinary lazy evaluation. Thus the extra OPEN cycle can erase a real reduction in heuristic work; a fixed prefix is useful only when it filters enough full evaluations. The contribution is this controlled empirical combination and mechanism decomposition, not a claim to invent landmarks, subsets, or multi-stage Lazy A*.

Introduction

Heuristics save search only after their own computational cost is paid. This tension is easy to hide when heuristic evaluation is treated as a constant: a more informed admissible estimate generally expands no more states under a fixed ordering, but it may still take longer to solve a problem. Landmark heuristics make the tradeoff concrete. A preprocessing pass stores exact distances from selected pivots; a query then obtains lower bounds from the triangle inequality. More pivots can strengthen the maximum, but each evaluated state performs more table reads.

Lazy A* changes when this cost is paid. A newly discovered state first receives a cheap bound. Only if it rises to the top of OPEN is an expensive bound computed and the state reinserted; surplus states may never receive the expensive

heuristic (Tolpin et al. 2013; Karpas et al. 2018). The saved heuristic calls compete with extra priority-queue operations. With a set of 32 differential landmarks, there is a natural intermediate design: compute a fixed four-landmark prefix at the first competitive pop, and compute the other 28 terms only if the state remains competitive at a later pop.

This is a deliberately bounded course project. It stays within optimal single-agent A*, admissible and consistent heuristics, OPEN-list behavior, and empirical comparison—the core material of a search-methods course. The extension adds no learned selector, calibration phase, rational bypass, multi-agent solver, or new domain. Its research question is narrow: *does an intermediate fixed landmark prefix save enough remaining heuristic work to justify another OPEN cycle, and in which map regimes?*

We answer that question under a prospectively fixed protocol. Four methods share one deterministic graph-search engine: Manhattan distance, eager evaluation of 32 landmarks, ordinary Lazy A* from Manhattan to all 32 landmarks, and progressive evaluation from Manhattan to four and then 32 landmarks. The prefix and full counts are fixed at $K = 4$ and $M = 32$; no development outcome selects them. Twelve checksum-pinned Moving AI maps define a map-disjoint split: four development maps authorize the unchanged run, and eight sealed maps supply the reported 800-query evaluation. Timing uses rotated blocks in which every method occurs twice in every order position.

The central result is a mechanism/performance split. The staged method did less deterministic landmark work: median pivot and read savings were 3.10% and 3.02% relative to ordinary lazy evaluation. Nevertheless, its median paired time ratio was 1.0543, with a map-hierarchical 95% bootstrap interval from 0.9890 to 1.1016. Savings and timing varied by topology. Both maze maps had substantial landmark savings and staged/lazy ratios below one; both warehouse maps had zero median savings and ratios above one. Across queries, read saving and log time ratio had descriptive Spearman correlation $\rho = -0.6406$, consistent with a crossover but not identifying a causal effect.

The paper contributes (i) a fully specified three-level evaluation schedule for a nested differential heuristic; (ii) a correctness argument and a trace-invariance gate that separates scheduling from search order; and (iii) a reproducible, map-disjoint decomposition of saved pivot work, extra OPEN

work, runtime, and preprocessing amortization. The novelty boundary is intentionally modest. Landmark lower bounds, differential heuristics, heuristic subsets, Lazy A*, and extension to more than two stages are prior ideas. A targeted literature search did not locate this exact fixed- prefix, node-level staging experiment; that is not an exhaustive priority claim.

Background and Related Work

Optimal heuristic search. A* orders search states by $f(n) = g(n) + h(n)$, where g is path cost and h estimates remaining cost (Hart, Nilsson, and Raphael 1968). An admissible heuristic never exceeds the true remaining cost; consistency additionally requires $h(n) \leq c(n, n') + h(n')$ for every edge. These properties and the details of graph search, reopening, and tie breaking matter to optimality and comparisons of expansion behavior (Dechter and Pearl 1985). Our implementation uses only lower-bound keys, supports reopening, and makes all final-stage ordering deterministic.

Landmarks and differential heuristics. ALT uses exact distances to selected landmarks with triangle inequalities to accelerate shortest-path search (Goldberg and Harrelson 2005). For an undirected graph, a differential heuristic takes $|d(\ell, n) - d(\ell, t)|$ and maximizes over pivots (Sturtevant et al. 2009). Landmark placement, count, memory, and preprocessing are therefore established design dimensions, not contributions of this project. Likewise, selecting a fixed subset from many admissible heuristics has been studied directly (Rayner, Sturtevant, and Bowling 2013). We lock a deterministic farthest-first order and evaluate its first four members as an intermediate stage; we do not optimize a subset on the reported maps.

Lazy and selective deployment. Lazy A* initially keys a generated state with a cheap heuristic and computes an expensive heuristic only when that state becomes competitive in OPEN. The start is fully evaluated, and a valid popped goal is accepted before unnecessary refinement (Tolpin et al. 2013). Rational Lazy A* may bypass a costly evaluation using estimated usefulness and cost; the archival treatment develops contextual rational deployment more broadly (Karpas et al. 2018). Selective Max instead learns which heuristic to evaluate (Domshlak, Karpas, and Markovitch 2010, 2012). Our schedule is neither rational nor learned: every non-goal state must reach the complete 32-pivot bound before expansion. The Lazy A* literature describes extension to multiple heuristic stages as straightforward. Consequently, adding a third stage alone is not an algorithmic novelty claim.

Dynamic-heuristic work gives a broader formal setting for refining estimates during optimal search (Christen et al. 2025). We use the simpler nested, consistent case. Each promotion preserves or raises the key, and every fully expanded state uses the same final heuristic. This structure lets us test a stronger invariant than cost equality: eager, ordinary lazy, and staged full-landmark modes should enumerate successors in exactly the same order.

Position of this study. Prior work establishes all ingredients separately: landmark preprocessing, differential max-

ima, subset selection, lazy evaluation, and multi-stage extension. Our residual contribution is empirical. We combine a fixed four-pivot prefix with a fixed 32-pivot final bound, then measure whether suffix calls avoided on surplus nodes outweigh the added requeues and pops. This mechanism-first scope resembles the course’s example projects, which is useful pedagogically, but algorithmic facts and novelty boundaries here are supported by the primary literature rather than by course examples.

Questions and Hypotheses

The study has four research questions.

RQ1: Correctness and control. Do all four modes return the independent BFS cost, and do the three modes that ultimately use H_{32} have identical successor-expansion traces?

RQ2: Mechanism. Relative to LAZY-FULL, how many pivot evaluations, distance-table reads, and suffix calls does STAGED avoid, and how many additional OPEN operations does it introduce?

RQ3: Search time. What is the paired staged/lazy search-time ratio over map-disjoint evaluation queries, with maps rather than timing repetitions treated as the top-level independent units?

RQ4: Regimes and amortization. How do the mechanism and timing ratio vary across maze, random, room, and warehouse maps, and how does one-time landmark construction affect total cost at different query volumes?

The prospectively encoded analysis maps these questions to three hypotheses. H1 is a hard gate: all 800 sealed queries must pass BFS equality and full-landmark trace invariance before timing is analyzed. H2 expects staged evaluation to save exact pivot evaluations and distance reads relative to ordinary lazy evaluation. H3 evaluates the paired time ratio and the association between saved work and time; it is explicitly descriptive. In particular, the bootstrap fraction of replicates with a ratio below one is not treated as a frequentist p -value, and the Spearman coefficient does not support a causal claim.

Search Model and Heuristics

Grid graph

Each map induces an undirected unit-cost graph $G = (V, E)$. A vertex is a cell marked “.”; “@” and “T” are blocked. Edges connect horizontal and vertical neighbors only. There are no diagonals or wait actions. A query has distinct traversable start s and goal t in the same connected component. Although scenario files contain a reference distance, it may reflect a different movement model and is ignored. An independent four-neighbor BFS supplies the oracle.

Nested differential bounds

Let $d(x, y)$ be exact graph distance and let ℓ_i be the i th map-level landmark. Manhattan distance is

$$H_0(v, t) = |x_v - x_t| + |y_v - y_t|. \quad (1)$$

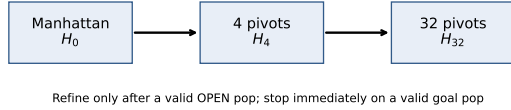


Figure 1: The proposed three-level staged-evaluation ladder. A non-goal STAGED state expands only after reaching H_{32} .

For one landmark, reverse triangle inequality gives

$$\phi_i(v, t) = |d(\ell_i, v) - d(\ell_i, t)| \leq d(v, t). \quad (2)$$

Distances involving a landmark in another connected component are ignored. For the first k pivots we define

$$H_k(v, t) = \max \left(H_0(v, t), \max_{1 \leq i \leq k} \phi_i(v, t) \right). \quad (3)$$

The experiment fixes $K = 4$ and $M = 32$, yielding

$$H_0(v, t) \leq H_4(v, t) \leq H_{32}(v, t) \leq d(v, t). \quad (4)$$

For adjacent v, w , the triangle inequality also gives $\phi_i(v, t) \leq 1 + \phi_i(w, t)$. Manhattan distance is consistent on the four-neighbor grid, and a pointwise maximum of consistent heuristics remains consistent. Thus every H_k is both admissible and consistent. Refining from H_0 to H_4 to H_{32} can only preserve or increase a state's lower-bound key.

Landmark order and storage

Placement is deterministic and independent of queries. Traversable cells are enumerated in row-major order; the first is ℓ_1 . Each later pivot maximizes its minimum distance to the selected set, with row-major tie breaking. An unrepresented connected component has infinite priority so it receives a representative before a represented component receives another, subject to the 32-pivot budget. One BFS per pivot produces immutable little-endian unsigned 32-bit distance rows with an explicit unreachable sentinel. The artifact records the ordered coordinates, map and table hashes, packed bytes, and build time.

Methods and Correctness

All conditions use one graph-search engine and differ only in when the nested heuristic is evaluated. Figure 1 shows the proposed staged ladder.

MANHATTAN. The start and every accepted state receive only H_0 and expand on a valid pop. This is the cheap, weak baseline.

EAGER-FULL. Every accepted state receives H_{32} immediately. This avoids heuristic promotion and requeue overhead but pays all 32 landmark terms even for states that never expand.

LAZY-FULL. Following Lazy A*, the start is fully evaluated before insertion. Other accepted states initially receive H_0 . A valid non-goal pop at the cheap stage computes H_{32} and requeues the state with the same insertion ordinal; a later full-stage pop expands it. A promotion occurs even when the numeric estimate does not increase.

STAGED. The start computes H_4 and then the remaining 28 terms before its initial insertion. Other states begin with H_0 . Their first valid non-goal pop computes H_4 and requeues; a second competitive pop computes only the remaining terms and requeues; a full-stage pop expands. This method cannot bypass the suffix if a state is to expand.

Shared graph-search contract

OPEN entries are ordered lexicographically by

$$(f, -g, ordinal, row-major\ id, version, stage). \quad (5)$$

A successful relaxation receives a fresh ordinal; heuristic promotion preserves it. Per-state labels store the best g , parent, version, ordinal, completed stage, and current h . Stale entries are discarded. Cached heuristic stages survive a later strict better- g relaxation because their values do not depend on g . CLOSED reopening is implemented even though consistency made its count zero in this experiment.

On a valid pop, the goal test precedes expensive refinement. Every heuristic is zero at the goal, so refinement could not raise its key. A non-goal state enumerates successors only at its condition's final stage. We define *expanded* accordingly; the returned goal is not counted as expanded and does not enter the expansion digest.

Optimality and trace equivalence

At every stage, $f = g + h$ is a lower bound on the cost of a solution through the entry. Suppose a goal with cost C is validly popped. If a cheaper solution of cost $C' < C$ existed, some frontier state on that path would have a current key at most C' . If unrefined, it would be popped, refined to another lower bound no greater than C' , and requeued; if fully refined, it would precede the goal. Either case contradicts the goal being minimal in OPEN. Finite maps, positive costs, stale-entry rejection, and reopening give completeness for every valid query.

Moreover, EAGER-FULL, LAZY-FULL, and STAGED all expand non-goal states only with the same H_{32} value. Promotion preserves ordinal, and Equation 5 fixes final-stage ordering. Therefore they should enumerate successors in the same sequence. The implementation hashes that sequence. Digest equality is used as a fail-closed empirical invariant in addition to the proof and independent BFS cost check.

Counters distinguish deterministic work from timing: expanded and generated states, relaxations, pops, stale pops, requeues, calls to Manhattan/prefix/suffix/full stages, pivot evaluations, table reads, cache hits, discovered states, and maximum OPEN/live-state sizes. Timing covers the complete search, while per-stage timers are disabled to avoid instrumentation changing the quantity measured.

Table 1: Frozen protocol. Each query-method pair has one warm-up and eight timed searches.

Split	Maps	Queries	Search runs
Development	4	160	5,760
Sealed evaluation	8	800	28,800
Total	12	960	34,560

Experimental Protocol

Prospective split and query selection

The checksum-pinned source snapshot contains Moving AI pathfinding maps and random scenario files (Sturtevant 2012). Table 1 summarizes the immutable plan. The same four topology families occur on both sides, but no map file overlaps. Development uses one train map per family and 40 queries per map. It validates correctness, completeness, binding, and the frozen analysis code; it performs no selection. Only after an external audit issues a freeze does sealed evaluation use two new maps per family and 100 queries per map.

For each map, scenario files 1–4 are read in original source order. A row is valid when endpoints are distinct traversable cells in the same four-neighbor component. The plan takes the first 10 valid rows per file for development and the first 25 for evaluation. Duplicate (map, s , t) keys fail closed. No query is sampled, replaced, or removed after timing.

The four development maps are maze-128-128-1, random-64-64-10, room-64-64-8, and warehouse-20-40-10-2-1. The evaluation’s validation-source maps are maze-128-128-2, random-32-32-20, room-64-64-16, and warehouse-10-20-10-2-1; its holdout-source maps are maze-128-128-10, random-32-32-10, room-32-32-4, and warehouse-10-20-10-2-2.

Balanced timing

Within every query, the base method order is Manhattan, eager full, lazy full, and staged. One untimed warm-up uses that order. Eight timed repetitions left-rotate it by repetition index, cycling through the four positions twice. Thus every method appears exactly twice in every position for every query. Runs are sequential on one machine (Windows 11, CPython 3.12.9, 8 logical CPUs); no concurrent search workers are used.

For a query and method, search time is the ordinary even median of eight observations: the mean of the fourth and fifth sorted values. Repetitions reduce measurement noise and are not statistical units. The primary staged/lazy estimand is

$$\exp\left(\text{median}_q \log \frac{T_{q,\text{staged}}}{T_{q,\text{lazy}}}\right). \quad (6)$$

Its percentile interval uses 10,000 deterministic bootstrap replicates (seed 23725513), sampling maps first and then queries within sampled maps. The eight maps are the top-level clusters. Quartiles use Hyndman–Fan type 7.

Gating and analysis

All 800 records must be present, all costs must match re-played BFS, every path must be legal, method summaries must match all repetitions, and the three full-landmark expansion digests must agree. Only then is timing unlocked. There are no post-hoc exclusions. Mechanism measures are exact per-query counter differences. Spearman associations use average ranks and are descriptive. Preprocessing amortization computes, within each map,

$$A_m(Q) = \bar{T}_m + B_m/Q, \quad (7)$$

where B_m is measured landmark construction time and Q is queries served; the reported value equal-weights the eight maps. Manhattan has $B_m = 0$; all landmark modes receive the same build charge.

Results

Integrity and hypothesis outcomes

All 800 sealed queries were retained. Independent BFS and deterministic search replay passed for every query; all four methods returned the BFS cost, and EAGER-FULL, LAZY-FULL, and STAGED had matching full-landmark expansion digests in all 800 cases. No post-hoc exclusion was made. These checks passed before performance fields were exposed to the analysis.

Table 2 reports the registered outcomes. H2 was supported: relative to LAZY-FULL, STAGED saved a median 3.0976% of pivot evaluations and 3.0172% of table reads. The corresponding query-level medians were 28 fewer pivots/reads and one avoided suffix call, while the median additional OPEN work was 116 combined queue operations (insertions and pops). Pivot-evaluation savings were nonnegative (420 positive, 380 zero) and skewed: the median was 3.10%, the mean 13.44%, and the IQR 0–27.34%. This is a heuristic-work improvement, not a wall-clock improvement.

Search time and deterministic work

The primary staged/lazy time ratio was 1.054300. Its map-then-query hierarchical bootstrap interval was [0.989043, 1.101585]; 5.3% of bootstrap replicates had a ratio below one. Since the interval includes one and only eight top-level clusters are available, we make no categorical dominance or significance claim.

Table 3 exposes the tradeoff. All full-landmark schedules averaged 160.69 expansions and 517.30 generated successors per query, compared with 1197.86 and 4043.00 for Manhattan. Eager evaluation minimized OPEN operations but averaged 9,945.68 pivot evaluations. Ordinary lazy evaluation reduced this to 8,353.88 at the cost of 260.06 requeues. Staging reduced it further to 6,048.29 but averaged 437.78 requeues and 599.86 pops. No method reopened a state. The mean-of-query-median times in Table 3 do not contradict the primary ratio: a ratio of paired times and a mean of raw times are different estimands, especially under strong map heterogeneity.

Table 2: Prospectively generated hypothesis outcomes. H3 is descriptive; its bootstrap fraction and correlation are not significance or causal tests.

Hypothesis	Status	Estimate	Interval
H1	passed	800/800 queries passed both gates	not applicable (exact validation)
H2	supported	pivot=0.0309758772; read=0.0301724138	descriptive IQRs in summary/map tables
H3	descriptive	ratio=1.05430023; Spearman rho=-0.640610853	hierarchical bootstrap 95% CI [0.989043253, 1.10158524]

Table 3: Overall per-query summaries on 800 evaluation queries. Time is the mean of each query’s eight-run median, in ms; counters are query means.

Method	Time	Expanded	Pops	Requeues	Reads
Manhattan	15.535	1197.86	1356.08	0.00	0.00
Eager full	8.149	160.69	162.09	0.00	9977.68
Lazy full	8.282	160.69	422.15	260.06	8385.88
Staged	8.056	160.69	599.86	437.78	6080.29

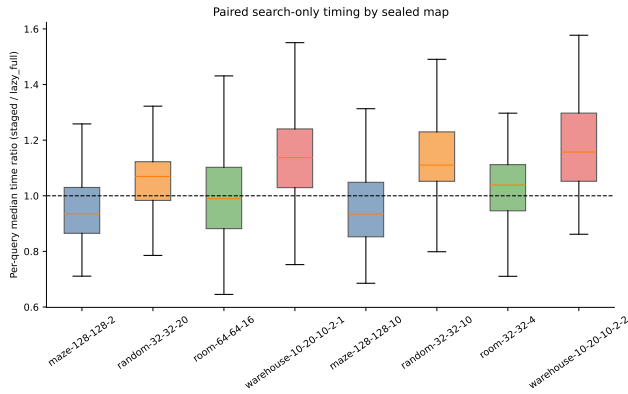


Figure 2: Per-map distributions of paired staged/lazy search-time ratios. The horizontal reference at one denotes equal time.

Map and family crossover

Figure 2 and Table 4 show repeatable family regimes. Both maze maps saved about 30–31% of reads and favored staging by about 6.5%. Both warehouse maps had zero median read saving and favored ordinary lazy evaluation by 13.7–15.7%. Random maps favored lazy, while the two room maps lay on opposite sides of one. The same direction on the two maze and two warehouse maps is suggestive replication within those families, not a population-level guarantee.

Across all 800 queries, read-saving fraction and log time ratio had Spearman $\rho = -0.640611$; extra OPEN work and log ratio had $\rho = -0.240994$. Figure 4 makes the crossover visible but does not isolate a causal coefficient: difficulty, topology, and absolute search size affect both the counters and time.

Preprocessing amortization

Preprocessing dominates when a table serves few queries (Figure 5, Table 5). At $Q = 100$ all landmark modes remain slower than Manhattan after charging construction; at $Q =$

Table 4: Per-map medians. Ratio < 1 favors staged evaluation; read saving is relative to ordinary lazy evaluation. V and H denote validation- and holdout-source maps.

Family	Source/map	Time ratio	Read saving
Maze	V: 128-2	0.935	31.17%
Maze	H: 128-10	0.934	30.34%
Random	V: 32-20	1.069	2.50%
Random	H: 32-10	1.110	0.00%
Room	V: 64-16	0.990	18.47%
Room	H: 32-4	1.039	11.46%
Warehouse	V: 10-20-1	1.137	0.00%
Warehouse	H: 10-20-2	1.157	0.00%

Table 5: Mean amortized ms/query, equal-weighting eight maps.

Q	Manhattan	Eager	Lazy	Staged
1	15.535	1881.862	1881.995	1881.769
10	15.535	195.520	195.653	195.427
100	15.535	26.886	27.019	26.793
1000	15.535	10.023	10.155	9.929

1000 they are faster under this equal-map average. Because all three landmark schedules share exactly the same table, preprocessing barely changes their relative order; it changes the practical comparison with Manhattan.

Discussion

RQ1: controlled correctness. The correctness and invariance gate passed without exceptions. Cost equality alone would allow subtle scheduling differences to remain hidden. Identical full-landmark expansion digests additionally show that eager, lazy, and staged conditions performed the same successor-enumerating search under the frozen tie breaker. Their differences in time and counters can therefore be localized to evaluation timing and OPEN promotions rather than to different expanded state sets.

RQ2: staging saved real work. The prefix was not merely ceremonial. Staging reduced median pivot and read work by about 3%, and much more on maze queries. In operational terms, LAZY-FULL evaluates all 32 terms whenever a cheap state becomes competitive; STAGED spends four reads first and avoids the other 28 when that prefix pushes the state below competitors. The saving is exact and deterministic. It is also heterogeneous and often zero, so the overall median understates large savings in one regime and overstates the effect in another.

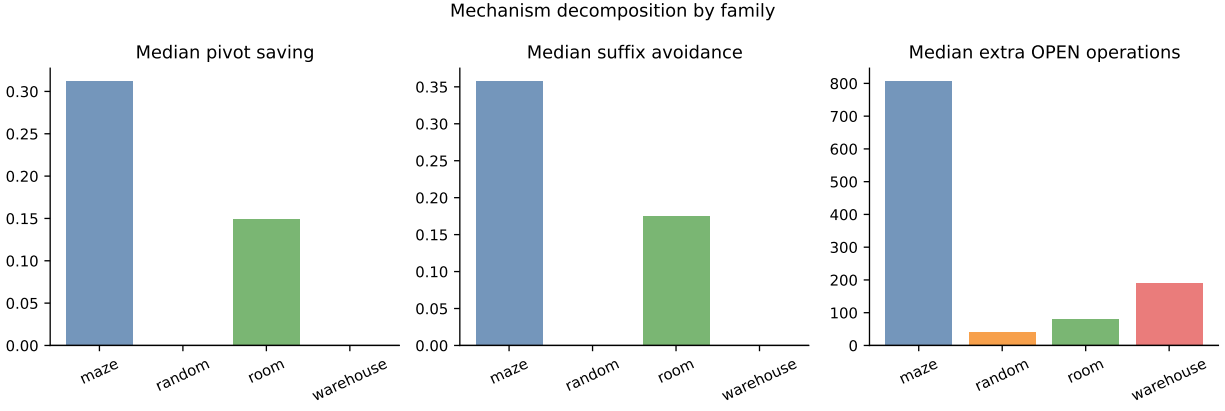


Figure 3: Family-level decomposition of saved landmark work and added OPEN work. Families are equal-weighted through their constituent queries.

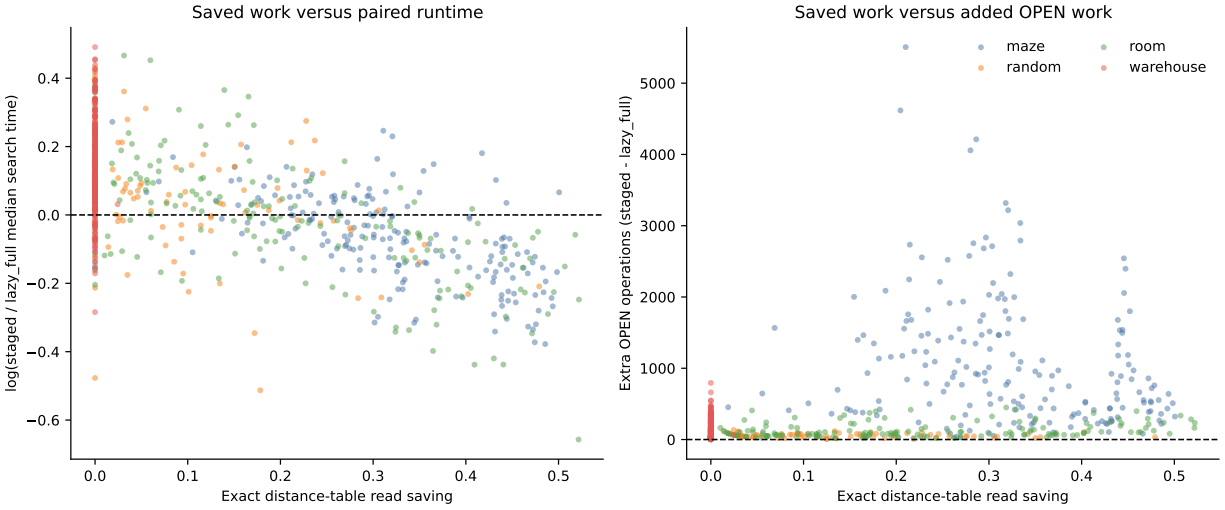


Figure 4: Query-level exact work changes versus paired time ratio. The association is descriptive and queries within a map are dependent.

RQ3: saved work did not imply universal speed. Every intermediate evaluation requires another pop and requeue. The staged schedule averaged roughly 178 more requeues and 178 more pops per query than ordinary lazy evaluation. Its primary paired ratio of 1.0543 therefore points toward a modest slowdown at the pooled median, while the 95% interval includes parity. We interpret this as bounded negative evidence for universal use of this fixed prefix, not as proof that staging can never help. Indeed, the maze maps show the expected positive regime.

The strong negative Spearman association between read saving and log time ratio supports the proposed mechanism descriptively: queries that avoided a larger fraction of reads tended to make staging relatively faster. It cannot establish that removing a particular read would cause the measured speedup. Both variables co-vary with topology, query length, absolute number of competitive nodes, cache locality, and

Python priority-queue costs.

RQ4: topology and workload matter. Warehouses are the clearest failure regime. The four-pivot prefix had zero median saving on both maps, so it added a nearly pure heap cycle before the same suffix work. Mazes are the clearest success regime: the prefix filtered enough candidates to save about 30% of reads on both maps, outweighing promotion overhead. Random and room maps occupy intermediate or mixed regimes. This crossover is more informative than a single leaderboard rank: it specifies when the simple schedule's mechanism is active.

Landmark construction changes the deployment decision at a different level. If a map receives only a few queries, even a fast landmark search cannot repay dozens of BFS preprocessing passes. At high query volume the fixed cost is small per query, and all landmark modes beat Manhattan under the study's equal-map average. The analysis deliberately charges

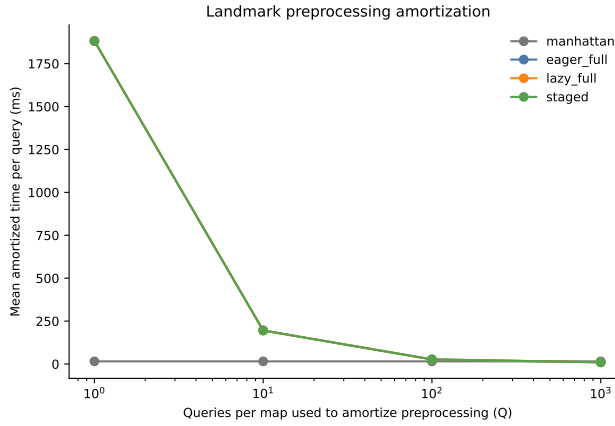


Figure 5: Equal-map-weighted amortized time as the one-time 32-landmark table construction is spread over Q queries.

the same build cost to eager, lazy, and staged modes because they use the same table; staging is a query-time schedule, not a preprocessing improvement.

Relation to prior work. The results instantiate the tradeoff identified by Lazy A*: heuristic calls saved on surplus nodes versus extra OPEN manipulation (Tolpin et al. 2013). They do not imply an optimal deployment policy and do not challenge rational or learned methods, which may choose conditionally whether to evaluate (Karpas et al. 2018; Domshlak, Karpas, and Markovitch 2012). A natural extension would use pre-decision features to choose between ordinary lazy and staged schedules. That is intentionally future work: fitting such a selector after seeing these maps would change the question and require a new split.

Limitations and Threats to Validity

Statistical scope. The evaluation has 800 queries but only eight top-level map clusters. The hierarchical bootstrap respects this structure, yet eight maps cannot support precise population claims. Two maps per family show useful within-family replication, not coverage of all instances in that family. The 0.053 bootstrap fraction is descriptive, not a p -value, and the interval is not evidence of equivalence.

External validity. All problems are undirected, unit-cost, four-neighbor grids from four known families. Results may change on weighted or directed graphs, road networks, diagonal movement, dynamic tasks, or other pivot-distance representations. Families occur on both sides of the map-disjoint split, so this tests new maps within known families rather than unseen-domain generalization.

Fixed design choices. $K = 4$, $M = 32$, and deterministic farthest-first placement are one locked configuration. The prefix may be unusually informative in some topologies and redundant in others. We did not compare alternate prefix sizes, pivot orders, compression, adaptive stopping, or active-landmark techniques. This restriction prevents tuning

to the sealed maps but limits the conclusions to the specified schedule.

Implementation and measurement. Packed-array reads and priority-queue operations have Python- and hardware-specific costs. Short searches remain sensitive to OS scheduling, cache state, allocator behavior, and timer resolution despite a warm-up, balanced order positions, and eight-run medians. Disabling stage timers avoids one perturbation but prevents assigning nanoseconds cleanly to prefix, suffix, and heap operations. Exact counters are more portable than run-time crossover points.

Internal validity. A shared engine and trace digest greatly reduce implementation confounds, but they also share any common bug. Independent BFS, legal-path validation, exhaustive small-map tests, immutable source hashes, deterministic replays, and strict artifact loaders provide independent layers. Expansion-trace equality is an implementation invariant under the frozen tie breaker; it is not a general theorem that every Lazy A* implementation must duplicate eager order.

Novelty. Every algorithmic ingredient has close prior art. ALT and differential heuristics establish landmark bounds; subset work establishes that a fixed selection can trade evaluation cost and informativeness; Lazy A* establishes delayed evaluation and notes straightforward multi-stage extension. We claim only a controlled study of their intersection and did not find an exact match in a targeted search. Differently named or unpublished work may exist.

Mechanical rerun. The first complete development/evaluation attempt is preserved separately. Before statistical output was published, analysis failed closed because it compared an immutable loaded schedule represented by tuples with the equivalent JSON plan represented by lists. One container-normalization fix and a direct regression test were added. No algorithm, protocol, query, K/M value, or estimand changed. Because the analysis source was prospectively hash-bound, all formal searches were rerun under the separately identified `progressive_landmarks_v2_rerun1` bundle. The 800 deterministic summaries match the first attempt exactly; only timings were remeasured. This is an analysis-integration correction, not a result-driven rerun.

Reproducibility and Provenance

Exact experiment and figure-generation code, configuration, source manifests, frozen raw results, and analysis are preserved at:

Experiment and figure-generation code:
<https://github.com/NevoLevi/progressive-landmark-astar/commit/4ee61db6787528efb7e01326e3c23d0006515570>

The authoritative configuration is `configs/progressive_landmarks_v2.json` (SHA-256 prefix 850176ee1990); its canonical plan hash begins 01aa82ec3984. The reported raw bundle is exclusively `data/results/progressive_landmarks_v2_rerun1`; the sealed manifest binds 800 query

records and 28,800 search runs. The analysis directory is `data/processed/progressive_landmarks_analysis_v2`; its manifest binds 800 query rows, eight map rows, summary/provenance files, and ten plots. The generated analysis summary hash begins `0f6e9fc457c3`.

The root `README.md` gives the exact copy-pasteable commands and arguments. Their fixed order is: (1) verify the protocol with `scripts/verify_progressive_landmarks_protocol.py`; (2) run development with `scripts/run_progressive_landmarks.py`; (3) audit and freeze it with `scripts/freeze_progressive_landmarks_development.py`; (4) invoke the same runner for sealed evaluation while supplying that freeze; and (5) generate all statistics and figures with `scripts/analyze_progressive_landmarks.py`. Every command binds the `v2` configuration and repository root. The final three stages also bind the development directory, audit, freeze, sealed directory, and analysis directory named above, so a superseded or substituted artifact fails closed.

Output directories are write-once and published atomically. Manifests bind file sizes, record counts, schemas, and SHA-256 hashes. Formal loading replays the canonical query plan, source maps, landmark tables, BFS oracle, paths, method summaries, counters, and expansion digests instead of trusting coherently edited stored fields. The development audit records `selection_performed=false`; its freeze is required to authorize sealed evaluation. Analysis again validates that authorization and all correctness gates before reading timing.

The report figures are byte-for-byte copies of the analysis PDFs. The hypothesis table is the generated $\text{\LaTeX}$ fragment rather than a manually retyped result. Compact method, map, and amortization tables were transcribed only from the immutable summary JSON and map CSV and are cross-checked in the report audit. Build the paper from `report/` with `pdflatex`, `bibtex`, and two final `pdflatex` passes.

Conclusion

Progressively evaluating a fixed differential-landmark prefix is correct and can avoid substantial heuristic work, but the additional OPEN cycle is not free. Under the locked $4 \rightarrow 32$ schedule, staging saved a median 3.10% of pivot evaluations and 3.02% of distance reads, while its primary paired time ratio relative to ordinary Lazy A* was 1.0543 with a 95% hierarchical bootstrap interval of $[0.9890, 1.1016]$. The pooled result does not support a universal speed advantage.

The map regimes answer the more useful question. A four-pivot prefix helped on mazes, where it removed about 30% of reads and both maps favored staging. It hurt on warehouses, where median read saving was zero and both maps favored ordinary lazy evaluation. Thus the relevant condition is whether the prefix filters enough full evaluations to repay its extra pops and requeues. That finding is descriptive and implementation-conditional, but it is supported by exact work counters, a negative read-saving/time-ratio association, independent BFS validation, and identical full-landmark expansion traces.

The study's scientific value lies in disciplined scope: it combines familiar course concepts into a reproducible experiment, states an honest prior-art boundary, preserves a sealed map-disjoint protocol, and reports a mixed result rather than tuning it away. Future work could preregister multiple prefix sizes or a deployment selector on new maps, port the engine to a lower-level language, and test weighted or directed graphs. Those extensions require new data and are not inferred from this evaluation.

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